

Summary

Adaptive Localization-Aware Goal-Directed Navigation for 2D LiDAR Mobile Robots

Anurag Karki

1 Goal-directed Exploration

Goal-directed exploration refers to a navigation scenario in which the robot knows its initial position and the goal location but does not have a prior map of the environment. Therefore, the robot must build the map online using SLAM while simultaneously navigating toward the specified goal.

Unlike conventional exploration, where the primary objective is to maximize coverage or gather information about the environment, goal-directed exploration must balance two objectives: exploring enough of the unknown environment to navigate safely and efficiently, while also making progress toward the final destination.

One important challenge in goal-directed exploration is localization uncertainty. As the robot moves through weakly observable or poorly localized regions, localization error can accumulate rapidly. If this uncertainty becomes too large, the robot may have difficulty accurately estimating its position, which can lead to poor planning, mapping errors, or even navigation failure. Recovering from severe localization degradation can also be difficult, particularly when the accumulated error is not immediately observable.

A possible solution is to identify weakly localized regions and encourage the robot to move toward areas that provide stronger localization information. However, continuously forcing the robot to remain in highly localizable regions may produce unnecessarily long or inefficient paths. The robot does not need to prioritize localization quality equally at every moment.

What is needed, therefore, is a mechanism that dynamically adjusts the importance of localization risk according to the robot's current state. In the proposed system, the robot's current pose covariance is used as feedback to regulate this behavior. When covariance is low, the planner can prioritize progress toward the goal and accept paths through moderately weak localization regions. As covariance increases, the planner progressively increases the importance of localization quality and favors paths through geometrically informative or strongly localizable areas.

In this way, the navigation system dynamically balances goal progress, exploration, and localization robustness based on the robot's current uncertainty and the information available in the partially constructed map. The objective is not to avoid every weakly localizable region, but to prevent the robot from entering such regions when its current localization state makes doing so excessively risky.